

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457630.678 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.031
Transit depth (Rp/Rs)^2	2.9e-05 +/- 0.0033 [%]
Orbital Inclination (inc)	88.58 +/- 2.79
Airmass coefficient 1 (a1)	1.099 +/- 0.02
Airmass coefficient 2 (a2)	-0.088 +/- 0.044
Scatter in the residuals of the lightcurve fit is	1.82 %
Best Comparison Star	#3 - [359, 217]
Optimal Aperture	6.4
Optimal Annulus	16.81
Transit Duration (day)	0.1118 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.031 vs 0.10538 (deviation 3.06σ)
Duration fit vs published	0.1118 d vs 0.1167 d (deviation 0.78σ)
Mid-time O–C	3.0 min
Depth SNR	0.0
Residual scatter	1.82 %

Light curve

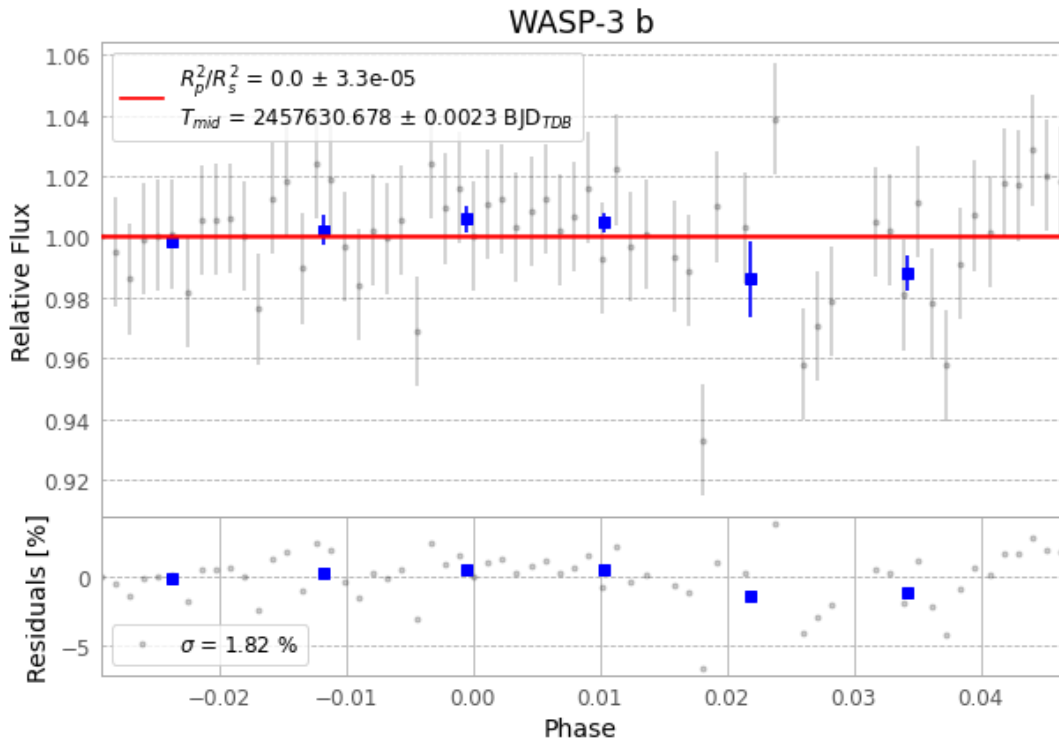


Figure 1 — FinalLightCurve_WASP-3 b_2016-08-29.png (SHA-256 e2f4b7ac8b19134f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [280, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 17d9fbb859684d92...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), WASP-3160830025412.FITS → WASP-3160830072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-160830.log (b4b75a352292...), FinalLightCurve_WASP-3 b_2016-08-29.png (e2f4b7ac8b19...), FinalLightCurve_WASP-3 b_2016-08-29.csv (3c2eedf7838a...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 04:42Z.